LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 1 Test (GSEB pattern): Chemical Reactions and Equations

Chapter / Topic: Chemical Reactions and Equations

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SECTION A

Q1. Which of the following is a correctly balanced chemical equation? **[1]**

Answer: (b) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$. Both sides have 3 Fe, 8 H and 4 O.

Q2. Which of the following is NOT a chemical change? **[1]**

Answer: (a) Melting of ice (only the state changes; no new substance is formed)

Q3. When dilute hydrochloric acid is added to zinc granules, the gas evolved is: **[1]**

Answer: (d) hydrogen. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Q4. In the reaction $2H_2S + SO_2 \rightarrow 3S + 2H_2O$, the substance that is oxidised is: **[1]**

Answer: (c) H_2S. It loses hydrogen.

Q5. $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ takes place in sunlight. This is an example of: **[1]**

Answer: (a) photolytic decomposition (decomposition by light)

Q6. Which of these is an endothermic process? **[1]**

Answer: (c) Decomposition of calcium carbonate (it needs heat)

Q7. In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, the substance that gets oxidised is: **[1]**

Answer: (b) H_2. It gains oxygen to form H_2O.

Q8. When lead nitrate is heated strongly, the products are: **[1]**

Answer: (d) lead oxide, nitrogen dioxide and oxygen. $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$

Q9. Which of the following is an exothermic process? **[1]**

Answer: (b) Respiration. Glucose is oxidised and energy is released.

SECTION B

Q10. Why should a magnesium ribbon be cleaned with sandpaper before burning it in air? Write the balanced equation for its burning. **[2]**

Answer: The ribbon has a layer of magnesium oxide (formed by reaction with air) on its surface, which stops it from burning easily. Sandpaper removes this layer. $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

- Reason (oxide layer removed): 1
- Balanced equation: 1

Q11. What is an oxidation reaction? Identify the substance oxidised in: $4Na + O_2 \rightarrow 2Na_2O$. **[2]**

Answer: A reaction in which a substance gains oxygen (or loses hydrogen). In $4Na + O_2 \rightarrow 2Na_2O$, sodium is oxidised, because it gains oxygen.

- Definition: 1
- Sodium oxidised, with reason: 1

Q12. What is a precipitation reaction? Give one example with a balanced chemical equation. **[2]**

Answer: A reaction in which an insoluble solid (precipitate) is formed. $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$, which gives a white precipitate of $BaSO_4$.

- Definition: 1
- Correct balanced example: 1

SECTION C

Q13. A solution of a substance 'X' is used for whitewashing walls. **[3]**

(i) Calcium oxide (quicklime), CaO [1]

(ii) $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + heat$ [1]

(iii) $Ca(OH)_2$ reacts slowly with CO_2 in air to form a thin, shiny layer of calcium carbonate: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ [1]

Q14. A shiny brown element X becomes black when heated in air. **[3]**

(i) X is copper (Cu). The black substance is copper(II) oxide (CuO). [1]

(ii) $2Cu + O_2 \rightarrow 2CuO$ [1]

(iii) Combination reaction (also an oxidation of copper). [1]

SECTION D

Q15. In a class activity, students heated lead nitrate powder in a boiling tube. Brown fumes were given off and a yellowish residue was left behind. In… **[4]**

(i) Nitrogen dioxide (NO_2). $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$ [1]

(ii) Thermal decomposition reaction [1]

(iii) Basic copper carbonate. Copper reacts slowly with moist carbon dioxide (and oxygen) in the air. This is corrosion of copper. [2]

- Names basic copper carbonate: 1
- Reason (reaction with moist CO_2/air, corrosion): 1